

## CHAMPAIGN WILLARD, IL, USA

WMO: 725315

Lat: 40.040N

Lon: 88.278W

Elev: 754

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94870

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.7       | 3.7 | -8.3                          | 3.6 | -0.5 | -3.0 | 4.8 | 5.3  | 31.8                  | 28.7 | 28.4 | 28.0 | 9.8                   | 260  | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 91.3            | 75.0 | 89.6 | 74.7 | 87.1 | 73.4 | 79.1                | 87.6 | 77.4 | 85.7 | 75.7 | 83.2 | 10.7                 | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.6                            | 142.7 | 84.8 | 74.8 | 134.2 | 82.5 | 73.2 | 126.7 | 80.4 | 43.1          | 87.8 | 41.3 | 85.5 | 39.7 | 83.3 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |       |  |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|-------|--|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |       |  |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |       |  |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |  |
| 27.6              | 24.6 | 21.4 | DB                         | -9.4 | 95.1 | 6.8                | 2.0                                                | -14.2     | 96.5 | -18.2      | 97.7 | -22.0      | 98.8 | -26.9      | 100.3 |  |
|                   |      |      | WB                         | -9.6 | 82.1 | 6.4                | 1.6                                                | -14.2     | 83.2 | -18.0      | 84.1 | -21.6      | 85.0 | -26.2      | 86.2  |  |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 52.3   | 25.6  | 30.1  | 40.6  | 52.7  | 63.6  | 72.2  | 74.3  | 72.4  | 67.1  | 54.8  | 41.7  | 30.9  |      |
|                                                                               | DBStd                                                                         | 19.55  | 12.44 | 11.67 | 11.14 | 9.50  | 8.76  | 6.29  | 5.30  | 5.00  | 7.46  | 9.60  | 10.05 | 10.90 |      |
|                                                                               | HDD50                                                                         | 2681   | 759   | 562   | 334   | 79    | 6     | 0     | 0     | 0     | 1     | 58    | 285   | 597   |      |
|                                                                               | HDD65                                                                         | 5655   | 1221  | 977   | 758   | 381   | 137   | 14    | 2     | 6     | 63    | 341   | 699   | 1056  |      |
|                                                                               | CDD50                                                                         | 3518   | 2     | 5     | 43    | 160   | 428   | 667   | 752   | 695   | 515   | 208   | 37    | 6     |      |
|                                                                               | CDD65                                                                         | 1019   | 0     | 0     | 2     | 12    | 94    | 232   | 290   | 236   | 127   | 25    | 1     | 0     |      |
|                                                                               | CDH74                                                                         | 9386   | 0     | 0     | 22    | 168   | 929   | 2144  | 2653  | 1973  | 1234  | 259   | 4     | 0     |      |
|                                                                               | CDH80                                                                         | 3232   | 0     | 0     | 2     | 30    | 309   | 791   | 956   | 626   | 452   | 66    | 0     | 0     |      |
| Wind                                                                          | WSAvg                                                                         | 10.3   | 12.6  | 12.3  | 12.4  | 13.1  | 11.4  | 9.1   | 6.9   | 6.3   | 7.3   | 9.9   | 11.5  | 11.6  |      |
| Precipitation                                                                 | PrecAvg                                                                       | 39.5   | 2.0   | 1.9   | 3.1   | 3.8   | 4.4   | 4.0   | 4.4   | 4.1   | 3.3   | 2.8   | 3.2   | 2.6   |      |
|                                                                               | PrecMax                                                                       | 58.5   | 6.2   | 4.5   | 7.8   | 9.6   | 10.1  | 9.1   | 13.8  | 10.0  | 8.7   | 7.2   | 10.1  | 6.6   |      |
|                                                                               | PrecMin                                                                       | 26.9   | 0.1   | 0.3   | 0.7   | 0.6   | 0.6   | 0.3   | 1.3   | 1.3   | 0.3   | 0.2   | 0.6   | 0.3   |      |
|                                                                               | PrecStd                                                                       | 5.8    | 1.4   | 1.0   | 1.7   | 2.1   | 2.3   | 2.2   | 2.8   | 2.3   | 2.0   | 1.6   | 1.9   | 1.6   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 60.4  | 64.3  | 77.2  | 83.6  | 91.0  | 93.4  | 95.0  | 92.4  | 93.0  | 86.3  | 72.6  | 63.1  |      |
|                                                                               |                                                                               | MCWB   | 56.6  | 55.9  | 62.9  | 64.3  | 70.3  | 74.0  | 78.8  | 76.8  | 72.7  | 69.6  | 59.1  | 59.0  |      |
|                                                                               | 2%                                                                            | DB     | 54.0  | 57.5  | 70.0  | 79.1  | 87.1  | 90.6  | 90.9  | 89.4  | 89.4  | 81.2  | 66.0  | 56.6  |      |
|                                                                               |                                                                               | MCWB   | 51.8  | 52.7  | 59.2  | 62.1  | 68.8  | 73.8  | 77.8  | 76.2  | 71.4  | 65.5  | 58.0  | 53.0  |      |
|                                                                               | 5%                                                                            | DB     | 48.0  | 52.2  | 63.6  | 74.1  | 83.6  | 88.2  | 88.4  | 86.2  | 85.3  | 75.7  | 62.2  | 51.3  |      |
|                                                                               |                                                                               | MCWB   | 45.2  | 47.8  | 55.3  | 60.5  | 67.1  | 73.3  | 76.8  | 74.7  | 69.9  | 62.8  | 56.2  | 48.5  |      |
|                                                                               | 10%                                                                           | DB     | 41.8  | 46.0  | 58.2  | 69.6  | 79.5  | 84.7  | 85.5  | 83.5  | 81.5  | 71.1  | 57.2  | 45.9  |      |
|                                                                               |                                                                               | MCWB   | 39.5  | 42.3  | 52.0  | 58.3  | 65.6  | 71.3  | 74.6  | 73.0  | 67.5  | 61.4  | 51.3  | 42.9  |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 57.2  | 59.1  | 64.4  | 68.4  | 74.4  | 79.1  | 82.1  | 80.5  | 76.9  | 72.0  | 63.5  | 60.1 |
|                                                                               |                                                                               |        | MCDB  | 59.5  | 63.0  | 74.7  | 79.0  | 84.5  | 87.9  | 90.4  | 88.5  | 87.2  | 82.6  | 67.7  | 62.9 |
| 2%                                                                            |                                                                               | WB     | 52.1  | 53.7  | 60.7  | 65.2  | 71.9  | 77.0  | 80.0  | 78.1  | 74.6  | 68.7  | 60.0  | 53.8  |      |
|                                                                               |                                                                               | MCDB   | 53.5  | 56.8  | 67.7  | 73.2  | 82.4  | 86.3  | 88.5  | 86.4  | 83.9  | 75.8  | 64.7  | 56.3  |      |
| 5%                                                                            |                                                                               | WB     | 45.7  | 47.8  | 57.3  | 62.6  | 69.9  | 75.3  | 78.2  | 76.2  | 72.4  | 65.9  | 56.8  | 48.2  |      |
|                                                                               |                                                                               | MCDB   | 47.6  | 51.8  | 62.8  | 70.5  | 79.0  | 84.1  | 86.2  | 83.6  | 80.5  | 72.7  | 61.4  | 50.4  |      |
| 10%                                                                           |                                                                               | WB     | 39.2  | 42.6  | 52.6  | 59.8  | 67.9  | 73.6  | 76.5  | 74.6  | 70.3  | 62.6  | 52.5  | 43.4  |      |
|                                                                               |                                                                               | MCDB   | 41.5  | 46.3  | 57.7  | 67.9  | 76.3  | 82.1  | 83.6  | 81.1  | 77.7  | 69.5  | 56.8  | 46.1  |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 15.4  | 16.2  | 19.1  | 22.1  | 21.3  | 20.0  | 19.3  | 19.9  | 23.3  | 21.9  | 18.1  | 14.2  |      |
|                                                                               |                                                                               | MCDBR  | 20.4  | 23.2  | 26.0  | 28.3  | 25.9  | 22.9  | 21.0  | 22.4  | 26.9  | 27.5  | 24.0  | 19.5  |      |
|                                                                               |                                                                               | MCWBR  | 17.7  | 18.5  | 16.6  | 15.2  | 11.4  | 10.0  | 10.7  | 11.4  | 11.5  | 13.9  | 16.5  | 16.6  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 19.9  | 22.2  | 23.6  | 23.8  | 22.0  | 19.1  | 19.0  | 19.2  | 21.9  | 22.6  | 21.6  | 18.1  |      |
|                                                                               |                                                                               | MCWBR  | 18.2  | 19.0  | 16.0  | 15.3  | 11.4  | 9.7   | 10.9  | 11.0  | 10.8  | 13.3  | 16.9  | 16.4  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.301  | 0.317 | 0.353 | 0.395 | 0.435 | 0.439 | 0.459 | 0.457 | 0.414 | 0.361 | 0.314 | 0.298 |       |      |
|                                                                               | taud                                                                          | 2.436  | 2.384 | 2.365 | 2.253 | 2.195 | 2.225 | 2.152 | 2.147 | 2.245 | 2.393 | 2.483 | 2.460 |       |      |
|                                                                               | Ebn at Noon                                                                   | 275    | 285   | 285   | 279   | 269   | 267   | 260   | 257   | 261   | 265   | 267   | 266   |       |      |
|                                                                               | Edh at Noon                                                                   | 25     | 31    | 35    | 42    | 46    | 44    | 47    | 46    | 39    | 30    | 24    | 23    |       |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 567    | 844   | 1158  | 1509  | 1723  | 1978  | 1951  | 1738  | 1465  | 1008  | 672   | 478   |       |      |
|                                                                               | RadStd                                                                        | 65     | 84    | 93    | 108   | 147   | 126   | 108   | 108   | 108   | 106   | 65    | 59    |       |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (46 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page